

Machine Learning Notes:

Instance-Based vs. Model-Based Learning

1. Introduction

In previous lectures, we categorized Machine Learning (ML) based on the amount of supervision (Supervised, Unsupervised, etc.) and how they update (Batch vs. Online).

Today, we categorize ML based on **how the model learns** (the learning process itself).

2. The Human Analogy: How do we learn?

The teacher explains that machine learning models learn exactly like human beings. There are two primary ways humans learn:

1. **Memorizing (Rote Learning / "Ratta"):** * This is when we simply remember things as they are without understanding the logic.
 - *Example:* In school, if we didn't understand a topic, we would memorize it for the exam.
2. **Generalizing (Understanding Concepts):**
 - This is when we try to understand the underlying principles or patterns behind the facts.
 - *Example:* In Physics, we try to understand the concept so we can solve any new problem based on that principle.

Machine Learning equivalents:

- Memorizing = **Instance-Based Learning**
- Generalizing = **Model-Based Learning**

3. Instance-Based Learning

Concept:

In this type of learning, the algorithm "memorizes" the training data. It doesn't extract any patterns during the training phase; it simply stores the data points and waits for a question (new data point).

The Placement Example:

Imagine we have a dataset with three columns:

1. **IQ**
2. **CGPA**

3. Placement (Target: Yes/No)

How Instance-Based Learning works:

- The model plots all the students' data on a graph (IQ vs. CGPA).
- Students who got placed are marked **Red**, and those who didn't are marked **Blue**.
- **The Learning Phase:** The model does nothing. It just "sits" with the data. It is often called a **Lazy Learner**.
- **The Prediction Phase:** When a new student (Green Point) comes with a specific IQ and CGPA:
 1. The model looks at the "Similarity."
 2. It calculates the distance between the new point and all existing points.
 3. It finds the "Nearest Neighbors" (e.g., the 3 closest students).
 4. If 2 out of 3 neighbors were placed (Red), the model predicts: **"Yes, this student will be placed."**

Key Logic:

"Where you live tells something about you." (Your neighbors define your category).

Example Algorithm:

- **k-Nearest Neighbors (kNN):** This is the classic example of Instance-Based Learning.

4. Model-Based Learning

Concept:

In this approach, the algorithm doesn't just store the data. It tries to find a **Mathematical Function** or a "Relationship" between the input and the output.

The Placement Example (Same Scenario):

- Instead of looking for neighbors, the model looks at the whole dataset and tries to draw a **Decision Boundary** (a line or curve).
- **The Learning Phase:** The model "trains" on the data to find the best possible line that separates the "Placed" group from the "Not Placed" group.
- **The Result:** Once the training is done, the model gets a mathematical formula (e.g., $y = mx + c$). This formula is the "Model."

The Prediction Phase:

- Once we have the line (Decision Boundary), we don't need the training data anymore!
- If a new point falls on the right side of the line $\rightarrow$ **Placed**.
- If it falls on the left side $\rightarrow$ **Not Placed**.

Example Algorithms:

- Linear Regression
- Logistic Regression
- Support Vector Machines (SVM)
- Neural Networks

5. Line-by-Line Logic Explanation (The Process)

For Instance-Based Learning:

1. **Store Data:** Save all IQ, CGPA, and Placement values.
2. **Wait for Query:** Do nothing until a new student's data arrives.
3. **Calculate Distance:** Find how "close" the new student is to every student in the database.
4. **Vote:** Look at the 'K' closest students and pick the majority result.

For Model-Based Learning:

1. **Analyze Data:** Look at the distribution of all points.
2. **Extract Rule:** Find a mathematical boundary (like a line) that separates the classes.
3. **Create Parameters:** Save the weights/slopes (parameters) of that line.
4. **Discard Data:** Delete the original training data (optional, as the rule is enough).
5. **Predict:** Plug the new student's values into the mathematical formula to get the answer.

6. Detailed Comparison Table

Feature	Model-Based Learning	Instance-Based Learning
Data Preprocessing	Required (Outlier removal, encoding).	Required (Outlier removal, encoding).
Core Goal	Create a "Model" (set of parameters).	Store the data in a suitable form.
Training Phase	Active (Model learns the parameters).	Passive (Just stores the data).
Storage Requirement	Low. Only needs to store the formula/parameters.	High. Needs to store the entire training dataset.
Generalization	Happens Before prediction (finds a general rule).	Happens During prediction (local to the new point).
Prediction Basis	Uses a mathematical "Decision Boundary".	Uses "Similarity" or distance to neighbors.

Data Dependency	Can discard training data after training.	Must keep training data forever.
------------------------	---	----------------------------------

7. Important Tips & Highlights

- **Lazy Learner:** Instance-based learning (like kNN) is called a "Lazy Learner" because it doesn't do any work until you ask it a question.
- **Storage Warning:** If you have 1GB of training data, an Instance-Based model will always take 1GB of storage on your server. A Model-Based algorithm might only take a few Kilobytes (the size of the mathematical formula).
- **Teacher's Pro-Tip:** Whenever you learn a new algorithm in the future (like Linear Regression or kNN), always ask yourself: **"Is this Instance-Based or Model-Based?"**
- **Outliers:** Both methods require clean data. For example, a student with low IQ but who got placed (due to luck/other factors) is an outlier and can confuse both models.

Summary of Terms

- **Decision Function:** The mathematical rule used in Model-Based learning to classify points.
- **Parameters:** Values (like weights or slopes) that define the Model.
- **Inference/Scoring:** The process of predicting the outcome for a new data point.